

Resonate: An AI-Powered Spotify Assistant

1. Project Overview

Resonate is a web application with an AI-Agent that has access to Spotify data on users. It has the power to identify a user's music taste, make song recommendations, and assess musical match between users. Its goal is to build off the platform on Spotify to better resonate with users.

2. Problem Statement & Motivation

Spotify stores lots of data on its users; however, those users are only given a small glimpse into their data once a year with Spotify Wrapped. Spotify Wrapped generates a lot of excitement every year which shows a strong desire for more interpretation of listening habits. Spotify users love music and they find the statistics behind their listening habits interesting. By adding a chatbot to that, Resonate can tap into a goldmine of data by providing a strongly desired service to users whenever they want it.

3. AI Strategy & GenAI Use Case

Through the use of an agent, API calls can be personally made by the user without them even realizing they are making them. This can allow for personal aggregation of data without the need for any technical background, which goes beyond the limitations of a traditional dashboard. With the data retrieved through these custom queries, the agent can assess the user's listening preferences and provide unique interpretations for each user.

4. Proposed Application Design

The project would have a traditional chat-like panel on the right side of the screen. This is where the user performs most of the interaction with the AI. Then with the rest of the screen there's an automatically generated dashboard that changes based on the results of the most recent query.

Spotify users will login to the website when they want to study recent listening habits. From there they can ask the AI questions about changes over time or patterns in their musical taste. This design allows users of any technical background to interact with their data on Spotify with the Resonate Agent acting as a proxy for the Spotify API.

5. Draft Technical & Development Plan

Resonate will be built on a React/TypeScript tech stack with a GenAI API key (say ChatGPT, or Gemini) and a Spotify API key for personal data retrieval. Depending on time, a Postgres database will be integrated with Docker to store messages and have them automatically

retrieved when a user signs in. While generative AI will be a large component of the application itself, it will also be very helpful with coding and assessing bugs where needed. In this regard it will help to fill in the gaps in my experience with web development acting as a UI designer and programming assistant.

6. Implementation Roadmap

Early Stages

Focus on core structure of the application and retrieving Spotify data.

- Establish a basic UI
- Utilize personal Spotify credentials for testing.
 - Run some Spotify API tests with the dashboard.
 - Hard code credentials for now.
- Build a simple chat interface capable of user input and AI output.

Middle Stages

- Construct agent logic to work with the Spotify API,
 - Deep studying and application of Spotify API logic.
 - Connect the results of the data retrieval to the AI for personalized interpretation.
- Add Spotify user authentication.

Late Stages

- Add agent logic to interact with dashboard
 - Use recent queries to populate dashboard
- Testing and website refinement

Extra Time

These are items to be implemented if time permits but technically lie outside of the scope of this course.

- Implement a database for message storage and retrieval
- Enable retrieval of monthly reports from the past